

Sugam Dahal

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SUMMARY

Computer Engineering graduate and machine learning engineer specialising in deep learning, computer vision and large language models. Industry experience deploying a production OCR microservice built on a vision-language model with vLLM, FastAPI and Docker, and evaluating post-training quantization for efficient LLM inference. Currently first-authoring a semantic segmentation study covering 202 km² of the Kathmandu Valley, spanning dataset construction, architecture design and evaluation of diffusion-generated training data. Comfortable across the full applied ML workflow: data pipelines, model training and fine-tuning, experiment design, statistical evaluation and deployment.

EDUCATION

Institute of Engineering (IOE), Thapathali Campus
Bachelor of Engineering in Computer Engineering | 78.56%
Global School of Science
Higher Secondary Education (Science) | GPA 3.59 / 4.00

Kathmandu, Nepal
Apr 2022 – Aug 2026
Kathmandu, Nepal
Jun 2019 – Dec 2021

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, Bash, LaTeX

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, Neural Networks, CNN, RNN, Transformers, Semantic Segmentation, Diffusion Models, Transfer Learning, Fine-Tuning, LoRA / PEFT, Model Quantization (GPTQ, AWQ), Mixed-Precision Training

LLMs & NLP: Large Language Models, GPT-2 Fine-Tuning, BPE Tokenization, Multi-Head Attention, Function Calling, Structured Outputs, Agentic Workflows, Prompt Engineering, LLM Evaluation (BLEU, Perplexity)

Computer Vision: Image Segmentation, Vision-Language Models, OCR, SegFormer, U-Net, Stable Diffusion, ControlNet, Albumentations, CVAT

MLOps & Deployment: FastAPI, Flask, vLLM, Docker, REST APIs, Git, GitHub, Linux, PostgreSQL, Weights & Biases, pytest, Pydantic, CUDA, Google Colab, Hugging Face Hub

Data Science & Statistics: NumPy, Pandas, Matplotlib, Seaborn, Data Preprocessing, Feature Engineering, PCA, Cross-Validation, Bootstrap Confidence Intervals, Experiment Design, Statistical Analysis, Data Visualization

PROFESSIONAL EXPERIENCE

Wise Yak Solutions Pvt. Ltd.
AI Research Intern

Kathmandu, Nepal
Dec 2025 – Mar 2026

- Engineered a production-ready OCR pipeline built on a 1B-parameter vision-language model, extracting structured text, dense tables and mathematical notation from complex document layouts.
- Architected a dual-engine inference backend using standard PyTorch for local development and vLLM with PagedAttention for high-throughput production serving, packaged behind a FastAPI service with Docker deployment.
- Evaluated post-training quantization methods (GPTQ, AWQ) for LLM inference, comparing memory footprint and output quality to inform deployment decisions on constrained hardware.
- Investigated custom neural network architectures and algorithm optimization within the AI Research Team, supporting applied computer vision research and deployment.

RESEARCH EXPERIENCE

Independent Researcher | Land-Use Segmentation and Generative Augmentation
Dept. of Electronics and Computer Engineering, IOE Thapathali Campus

Kathmandu, Nepal
May 2026 – Present

- Built and released a six-class land-use segmentation benchmark covering 202.4 km² of the Kathmandu Valley across 408 hand-annotated fine-resolution scenes, published on Zenodo with fold definitions, metric records and trained weights (DOI: 10.5281/zenodo.22349064).

- Designed and validated a full evaluation protocol across five architectures (U-Net, UNetFormer, SegFormer-B0, SegNeXt-T, proposed) and two data partitioning regimes, quantifying that split protocol alone shifts reported mIoU by 0.051–0.095, an effect larger than any architectural difference.
- Developed a thin-structure-aware segmentation model adding a residual detail path, boundary-reliability reweighting, a cDice topology loss and a presence-balanced sampler to a SegFormer-B0 backbone, improving on the backbone in every fold-run for 1.9% additional parameters.
- Implemented a two-stage diffusion generator (mask DDPM plus LoRA-adapted Stable Diffusion with a class-channel ControlNet) and a compute-matched downstream evaluation with bootstrap confidence intervals; first-author manuscript in preparation, preprint forthcoming.

PROJECTS

GeoNUSAF: GeoSpatial Neural Urban Synthesis and Analysis Framework

PyTorch, SegFormer, Stable Diffusion, ControlNet, FastAPI, Weights & Biases | Team of 4

- Fine-tuned a SegFormer-B0 backbone for 7-class urban land-use segmentation on 512x512 satellite imagery, using class-weighted cross-entropy for imbalance, albumentations-based augmentation and 3-fold cross-validation, reaching 0.50–0.60 validation mIoU with experiment tracking in Weights & Biases.
- Built a generative urban-planning pipeline feeding segmentation masks and templated prompts into Stable Diffusion with Canny-edge ControlNet conditioning, decoupling structural guidance via a reference image, and served the system through a FastAPI backend with automated cost and time estimation.

High-Performance OCR Microservice

Python, LightOnOCR-2-1B (VLM), vLLM, FastAPI, Docker, Hugging Face Transformers

- Built an end-to-end OCR microservice around a 1B-parameter vision-language model, replacing brittle multi-stage OCR pipelines with direct image-to-text inference over PDFs, scans and images.
- Implemented configurable dual-backend inference (Hugging Face for low-VRAM development, vLLM for production throughput), aspect-preserving 200 DPI PDF preprocessing, health and diagnostic endpoints, and containerised deployment via Docker Compose.

StockSage: Multi-Agent AI Financial Research Platform (Ongoing)

Python, Groq (Llama-3.3-70B), Pydantic, Pandas, yfinance, Tavily, pytest

- Developed an agentic research pipeline chaining real-time market fundamentals, 6-month OHLCV history, technical indicators (Wilder's RSI-14, SMA-20/50 implemented from scratch) and filtered news retrieval into a single per-ticker LLM summary.
- Implemented a generic ToolDispatcher with an agentic loop resolving multi-tool chains automatically, Pydantic-validated structured outputs via function calling, streaming with exponential-backoff retries, and a unit plus integration test suite across five modules.

Building Large Language Models from Scratch

PyTorch, Transformers, Multi-Head Attention, BPE Tokenization, tiktoken

- Implemented a complete GPT-style language model from first principles across 11 progressive modules: BPE tokenization, sliding-window data loaders, token and positional embeddings, multi-head self-attention, causal masking, layer normalisation and cross-entropy training with perplexity tracking.
- Engineered a weight-mapping pipeline converting pretrained OpenAI GPT-2 (124M) TensorFlow checkpoints into the custom PyTorch architecture for verified text generation and inference.

Interactive Chatbot for Mental Wellbeing

Python, PyTorch, GPT-2 Fine-Tuning, Flask, PostgreSQL, React | Team of 4

- Fine-tuned GPT-2 on 6 domain-specific Hugging Face mental health datasets over 10 epochs (batch size 4, learning rate 3e-5) to generate contextually aware supportive responses, evaluating generation quality with BLEU score.
- Built the Flask REST API and a four-table PostgreSQL schema handling authentication, interaction logging, ratings and feedback capture, integrated with a React frontend for real-time conversation.

Network Intrusion Detection System (NSL-KDD)

Python, Scikit-learn, XGBoost, Pandas, PCA

- Built an automated ML pipeline for network intrusion detection featuring programmatic Kaggle dataset extraction, categorical encoding, robust feature scaling and optional PCA dimensionality reduction.
- Benchmarked 7 classifiers (XGBoost, Random Forest, Linear SVM, KNN, Naive Bayes, Decision Tree, Logistic Regression) on accuracy, precision and recall with automated confusion-matrix and feature-importance visualisations.

CERTIFICATIONS

The Complete Full-Stack Web Development Bootcamp – Udemy, 2024

Hardware Fellowship Program – Locus, 2023

RESEARCH INTERESTS

Deep learning with a focus on computer vision, large language models, mechanistic interpretability and representation learning, bridging rigorous empirical methodology with deployable intelligent systems.